

AI-Powered Virtual Try-On System for Raritone

Final Internship Documentation & Recommendations Report

Executive Summary

This project researched AI-powered Virtual Try-On (VTO) solutions for fashion e-commerce. The objective was to analyze technologies that help customers visualize garments digitally before purchase, improving confidence, engagement, and conversion rates.

Project Objectives

Research AI/ML technologies for virtual try-on, evaluate datasets and models, study recommendation systems, identify implementation challenges, and propose a practical roadmap for Raritone.

Research Completed

The research covered Human Pose Estimation, Human Parsing, Image Segmentation, Cloth Warping, Garment Fitting, Recommendation Systems, Augmented Reality (AR), Virtual Reality (VR), and 2D/3D Virtual Try-On architectures.

Models Explored

MediaPipe, OpenPose, Detectron2, U2Net, Mask R-CNN, DensePose, VITON, CP-VTON, and Stable Diffusion were analyzed based on speed, accuracy, scalability, and suitability.

Datasets Collected

DeepFashion, DeepFashion2, VITON, VITON-HD, MPV, ModaNet, and Fashion-MNIST datasets were studied for pose estimation, segmentation, garment fitting, and recommendation tasks.

2D Virtual Try-On Findings

2D virtual try-on systems are easier to implement, cost-effective, and suitable for web applications. They provide fast results but have limitations in realism and depth perception.

3D Virtual Try-On Findings

3D systems provide realistic garment fitting, avatar generation, cloth simulation, and immersive shopping experiences. However, they require advanced hardware and development resources.

AI Recommendation System

A Hybrid Recommendation System combining Content-Based Filtering, Collaborative Filtering, and Deep Learning was identified as the most effective approach for personalization.

Challenges and Solutions

Major challenges included pose estimation errors, garment alignment, dataset inconsistencies, and real-time performance. Solutions included MediaPipe, DensePose, U2Net, TPS, and GMM techniques.

Complete AI Workflow

User Upload → Background Removal → Human Parsing → Pose Estimation → Garment Segmentation → Cloth Warping → Virtual Try-On Generation → Recommendation Engine → Final Output.

Suggestions for Raritone

Implement a 2D Virtual Try-On prototype first, integrate AI-based size recommendations, deploy a hybrid recommendation engine, and collect user feedback for continuous improvement.

Business Solutions for Raritone

Reduce return rates with AI size prediction, increase sales through personalized recommendations, improve customer confidence using virtual try-on, and leverage analytics for trend forecasting.

Future Improvement Suggestions

Real-time webcam try-on, mobile app deployment, AI fashion assistants, AR smart mirrors, virtual avatars, 3D body reconstruction, and cloud-based scaling.

Implementation Roadmap

Phase 1: 2D Virtual Try-On and Recommendations. Phase 2: Mobile Integration and Live Try-On. Phase 3: Full 3D AR/VR Experience and AI Styling Assistant.

Conclusion

The research demonstrates that AI-powered Virtual Try-On systems can significantly improve customer experience, engagement, personalization, and conversion rates. A phased implementation strategy is recommended for Raritone.